



Microstructural and hardness investigations on a dissimilar metal weld between low alloy steel and Alloy 82 weld metal

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ABSTRACT

The investigation on microstructure and hardness at the fusion boundary (FB) region of a dissimilar metal weld (DMW) between low alloy steel (LAS) A508-III and Alloy 82 weld metal (WM) was carried out. The results indicated that there were two kinds of FBs, martensite FB and sharp FB, with obvious different microstructures, alternately distributed in the same FB. The martensite FB region had a gradual change of elemental concentration across FB, columnar WM grains with high length/width ratios, a thick martensite layer and a wide heat affected zone (HAZ) with large prior austenite grains. By comparison, the sharp FB region had a relatively sharp change of elemental concentration across the FB, WM grains with low length/width ratios and a narrow HAZ with smaller prior austenite grains. The martensite possessed a K-S orientation relationship with WM grains, while no orientation relationship was found between the HAZ grains and WM grains at the sharp FB. Compared with sharp FB there were much more $\Sigma 3$ boundaries in the HAZ beside martensite FB. The hardness maximum of the martensite FB was much higher than that of the sharp FB, which was attributed to the martensite layer at the martensite FB.

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1. Introduction

In nuclear power plants, weld joints between low alloy steel (LAS) and nickel-base alloy are called dissimilar metal welds (DMWs) as LAS and nickel-base alloy are different kind of metals in composition, microstructure and crystallographic structure. Nickel-base alloy such as Alloy 152/52 and Alloy 182/82 are usually used in the DMW. The cracking incident of the DMW in V.C Summer nuclear power plant in 2000 raised concerns about the integrity of the DMWs [1]. A high strength mismatch which would lead to failures of the DMWs was found in the DMW between the low alloy steel and nickel-base alloy weld metal [2–4]. A micro-hardness peak, which could always be found at the fusion boundary (FB) region between the low alloy steel and the nickel-base weld metal, was believed to result from residual stress, precipitation and martensite formed during welding [5–11]. As a result, cracks had been found in the FB region or the locations adjacent to it [12–14]. Results of microstructure observations of DMWs between LAS and nickel-base alloy in former studies are different: on one hand, several studies showed that martensite might exist at the FB region in DMWs between LAS and the nickel-base weld metal [9,15,16]. Chung et al. [9] showed

the presence of a hard “martensite-like” layer which was responsible for the sharp change of the mechanical properties across the FB region by scanning electron microscope (SEM), Parker et al. [16] also showed an SEM image of a “lath-like” structure near the FB in a DMW between 2.25Cr-1Mo steel and Inconel 182 weld metal, while Olden et al. [15] showed a transmission electron microscope (TEM) image of a “twin-like martensite” structure in a DMW between AISI 8630H steel and Inconel 625 weld metal. However, the crystalline structure of all the “martensite-like”, “lath-like” and “twin-like” phases which might be martensite were not determined using any diffraction method in these studies. Pan et al. [17–19] suggested that the formation of the “martensite-like” structure at the FB region was resulted from alloy element distribution determined by convection of the molten metal in molten pool, stirring effect of mechanical force during welding and diffusion process of alloy elements during welding cycle. On the other hand, TEM observations of recent studies didn't show any existence of martensite layers along the FB [5,6,10,11]. These authors mentioned that there were lath structures near the FB in the heat affected zone (HAZ) in the LAS side, but they probably are the lath ferrite of the bainitic microstructure.

As the nickel-base alloy weld metals are very susceptible to stress corrosion crack (SCC) in nuclear power plants and the cracks are inclined to extend toward the FB, the microstructure of the FB region become important because microstructure is one of the key factor affecting the propagation behavior of SCC cracks. Since

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Table 1

Chemical composition of the base metal and filler metal in wt.%.

	C	Si	Mn	S	P	Cr	Ni	Mo	Ti	Nb	Fe
A508-III	0.22	0.24	1.60	0.003	0.008	0.1	0.72	0.55	–	–	Bal.
Alloy 82	0.04	0.05	2.80	0.001	0.001	21.2	Bal.	–	0.25	2.80	2.61

the inconsistency in the microstructure of the FB region in former studies, it is necessary to make a careful study on the microstructure observation of the FB regions which can help to understand the reason of the formation of the microstructure.

2. Experimental

A DMW between LAS A508-III and nickel-base WM Alloy 82, which was used to connect the reactor pressure vessel and the stainless steel safe end, was studied in this work. The DMW was prepared by multi-pass shielded metal arc welding. The welding current and voltage are 130 A and 25 V respectively. The welding speed is 17.6 cm/min and the heat input is 11.1 kJ/cm. The interpass temperature is 150 °C. The chemical composition of the base metal and the filler metal are listed in Table 1. The schematic of the weld joint is shown in Fig. 1 (a).

An area about 7 mm in length of the FB region, which was between two welding beads, was chosen for microstructure observation by optical microscope (OM), as shown in Fig. 1 (b). For OM microstructure observation, the weld metal side was electrochemically etched by 80% H₃PO₄, and then the LAS side was etched by 4% nital. Meanwhile, both SEM and TEM observations were carried out for further understanding the microstructure of the FB region. For SEM observation and electron backscattered diffraction (EBSD) scanning, a Hitachi SU-70 (field emission scanning electron microscope) FESEM equipped with a TSL Laboratories Orientation Imaging Microscope system at an accelerate voltage of 20 kV was used. For TEM observation, A Tecnai F30 TEM at an accelerate voltage of 200 kV was used. The TEM specimens were prepared using a Hitachi FB200A Focused Ion Beam system.

A Fischer MESSKAMMER HM500 nano indentation was used for measure the hardness within an area which is 250 μm in width across the FB region. The load and holding time used for the nano indentation measurement were 5 g and 10 s, respectively.

3. Results

Fig. 2 shows the OM microstructure of the FB in the 7 mm-area shown in Fig. 1 and some enlarged images of typical microstructure are inserted. The observation area includes the upper part of one welding bead and the bottom part of the following welding bead, as shown in Fig. 2 (a). The left side is Alloy 82 WM and the right side is a part of the heat affected zone (HAZ) of A508-III LAS. The enlarged images of the typical microstructures are shown in Fig. 2 (b)–(g). In these enlarged images, Alloy 82 side exhibits a columnar grain microstructure with carbides inside the grains, while the HAZ in the LAS side exhibits coarse prior austenite consisting of bainitic grains. On the other hand, in the enlarged images, there are two different types alternately distributed along the whole FBs. One shows dark-etched bands with different widths, as shown in Fig. 2 (c)–(f), while the other one exhibits a sharp boundary without the dark-etched band, as shown in Fig. 2 (b) and (g). The black and white lines close to the whole FB stand for the ratio of the FB with dark-etched band (black) and sharp (white) FBs in the whole FB, indicating the FB with dark-etched band possesses the most part of the whole FB, as shown in Fig. 2 (a).

Fig. 3 (a) and (b) show microstructures of both the WM sides and the LAS sides along typical dark-etched band FB and sharp FB, respectively. They were taken from the regions in the blue rectangles marked in Fig. 2. As shown in Fig. 3 (a), it is found that, in the HAZ beside dark-etched band FB, the prior austenite exhibits regular polygonal grain with a maximum diameter of about 100 μm. The dark-etched band with a thickness of about 20 μm grows perpendicularly to the FB from LAS side to WM side. By comparison, beside the sharp FB, as shown in Fig. 3 (b), it is found that in the HAZ, the prior austenite grains exhibits irregular smaller grains with a size of about 50 μm.

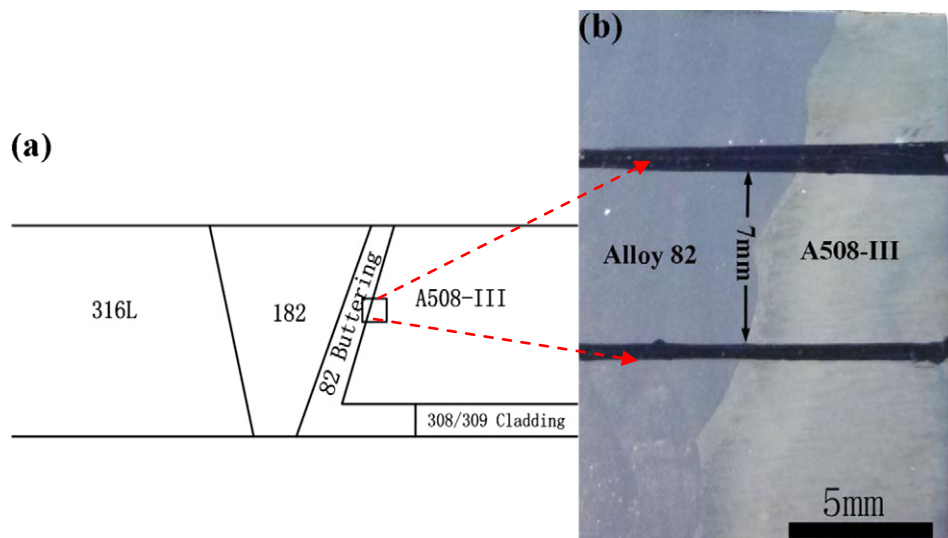


Fig. 1. (a) Schematic of the DMW joint, and (b) OM image of the chosen area in the DMW for observation.

Fig. 4 (a) and (b) show SEM images and the energy dispersive spectrometer (EDS) results across dark-etched band FB and sharp FB, respectively. The EDS measurement was carried out at the white dots on the white line marked in both images. It is found that beside the dark-etched band FB, the HAZ exhibits lath ferrite grains, while the dark-etched band exhibits fine acicular phases, as shown in Fig. 4 (a) [20]. Former studies believe these acicular phases are “martensite” as they found the phases are much harder than the surrounding phases. For this “martensite” FB, the EDS result shows that, from LAS side to WM side along the white line, iron concentration gradually decreases from 98% to about 60% and nickel concentration gradually rises from about 5% to about 30%. The changes of elemental concentrations show an approximately linear trend. By comparison, beside sharp FB, the HAZ exhibits small equiaxial grains with a diameter of 2–5 μm , as shown in Fig. 4 (b). For the sharp FB, the EDS result shows that, from LAS side to WM side along the white line, iron concentration rapidly decreases from 98% to about 30% and nickel concentration rapidly rises from about 0% to about 50% across fusion boundary. The elemental concentrations show a sudden change across the sharp FB.

Fig. 5 (a) is the IPF image showing microstructure characterizations of “martensite” FB region. As shown in Fig. 5 (a), it is found that the WM exhibits columnar grains which grow perpendicularly to the FB. Most of the WM grains are longer than 200 μm in length along the growing direction; however their widths are variable in the range of 30–100 μm . The length/width ratios of these WM grains are larger than 1. Fig. 5 (b) is an enlarged IPF image of the area in the black rectangular marked in Fig. 5 (a). For “martensite” FB, it is found that the “martensite” exhibits same structure and crystallographic orientation with the HAZ

grains as shown in Fig. 5 (b). The HAZ grains exhibit almost the same crystallographic orientations as they may belong to one prior austenite grain.

Fig. 5 (c) is the inverse pole figure (IPF) image showing microstructure characterizations of sharp FB region. By comparison, as shown in Fig. 5 (c), for sharp FB, some WM grains grow perpendicularly to the FB and only several of them are about 200 μm in length along the growing direction. Other WM grains grow randomly and the length/width ratios of these grains are less than or equal to about 1. Fig. 5 (d) is an enlarged IPF image of the area in the black rectangular marked in Fig. 5 (c). For sharp FB, the boundary between the WM and the HAZ is a smooth boundary without the existence of the “martensite” as shown in Fig. 5 (d). It is found that the HAZ grains beside sharp FB exhibit various crystallographic orientations as they may belong to different prior austenite grains.

Fig. 6 (a) and (b) show the distribution of coincidence site lattice (CSL) boundaries in the HAZ beside “martensite” FB and the sharp FB, respectively. Black line boundaries represent random grain boundaries (RGBs) and red ones represent CSL boundaries. From Fig. 6 (a), it is found that CSL boundaries distributed in most of the region in the HAZ beside “martensite” FB. By comparison, most of the CSL boundaries distributed in a region which is in a distance within 1 μm from the FB in the HAZ beside sharp FB, as shown in Fig. 6 (b). Fig. 6 (c) and (d) showed the statistic results of the CSL boundaries number fraction. It is found that most of the CSL boundaries in both conditions are $\Sigma 3$ boundary. The content of $\Sigma 3$ boundary in the HAZ beside “martensite” FB is 18% as shown in Fig. 6 (c), which is

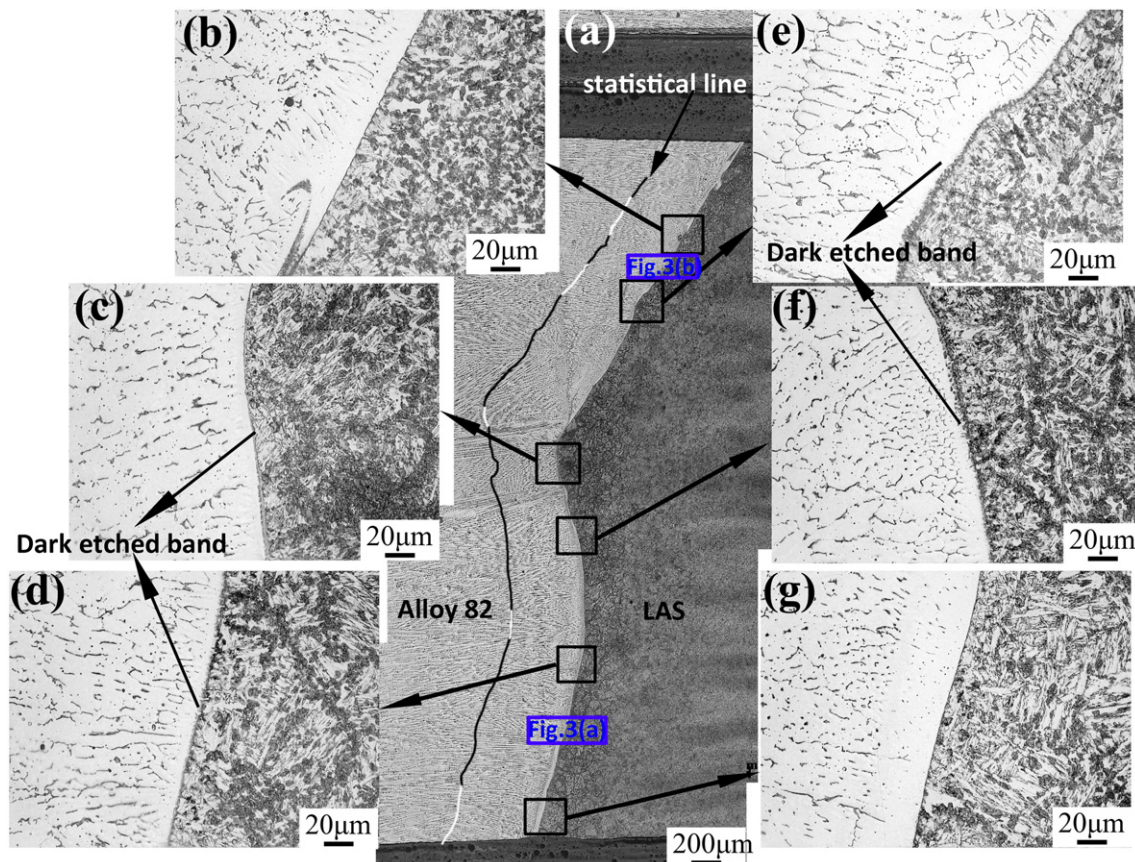


Fig. 2. (a) OM image of microstructure of the 7 mm-area in Fig. 1, and (b)–(g) high magnification images of different parts in panel (a).

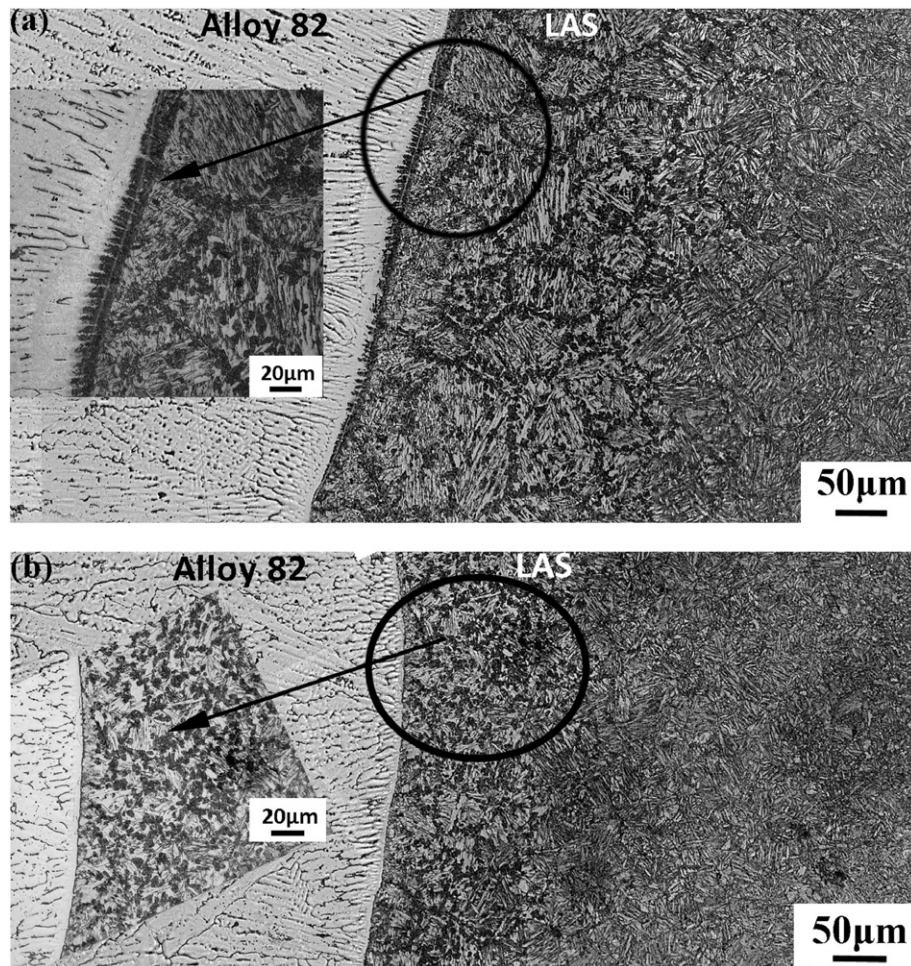


Fig. 3. OM images of microstructures of both the weld metal side and LAS side near (a) dark-etched band and (b) sharp FBs.

much more than that of $\Sigma 3$ boundary in the HAZ beside sharp FB ($<4\%$), as shown in Fig. 6 (d). For the HAZ beside “martensite” FB, there are also small amount of $\Sigma 9$ ($<0.01\%$), $\Sigma 11$ (about 4%), $\Sigma 13b$ ($<0.01\%$) and $\Sigma 25b$ ($<0.01\%$) boundaries. For the HAZ beside sharp FB, small amount of $\Sigma 5$ ($<0.01\%$), $\Sigma 7$ ($<0.01\%$), $\Sigma 9$ ($<0.01\%$), $\Sigma 11$ ($<0.01\%$), $\Sigma 13b$ ($<0.02\%$) and $\Sigma 17b$ ($<0.01\%$) boundaries are found.

Fig. 7 (a) shows the TEM image of the “martensite” FB, and the selection area electron diffraction (SAED) patterns of the areas “1” and “2” marked in Fig. 7 (a) are inserted, respectively. It is found that the WM exhibits F.C.C structure and HAZ exhibits body center cubic (B.C.C) structure. A lenticular phase is found at the “martensite” FB, as shown in Fig. 7 (a). Fig. 7 (b) shows the SAED pattern of the area “3” marked in Fig. 7 (a). As presented by the SAED patterns, the lenticular phase with B.C.C structure exhibits a K-S orientation relationship with the (face center cubic) F.C.C phase, indicating shear transformation takes place in the F.C.C WM [22]. Therefore, the dark-etched band is a martensite layer along the martensite boundary. By comparison, the boundary between WM and HAZ is a smooth boundary and there is no defined orientation relationship, as shown in Fig. 7 (c).

Fig. 8 shows the hardness profiles measured by nano indentation across martensite FB and sharp FB. It is found that the area around the sharp FB is generally harder than the area around

the martensite FB, which is believed to be resulted from the residual strain due to sharp changes of the microstructure and chemical composition across sharp FB. As shown in Fig. 8, except the hard peak at the FB, the average hardness value of the region around martensite FB is $Hv_{0.005}291.8$; while for sharp FB, the average hardness value of is $Hv_{0.005}313.5$. On the other hand, the hardness maximum is located at the FB in both hardness profiles. The hardness maximum of the martensite FB ($Hv_{0.005}497.6$) is much higher than that of the sharp FB ($Hv_{0.005}348.3$), which is resulted from the martensite layer at the martensite FB.

4. Discussion

Two kinds of FBs, sharp FB and martensite FB, are found in the DWM between LAS A508-III and nickel-base WM Alloy 82, as shown in Fig. 2. Up to now, most former studies on DMW between LAS and nickel-base WM seldom focused on the microstructure of FB. In our study, the sharp FBs, which were characterized in Figs. 2 (b) and (g) and 3 (b) and 4 (b), possessed about 1/4 of the whole FB. The sharp FB was usually found in the DMWs between LAS A533B/A508-III and nickel-based WM Alloy 182 (or Alloy 152) [5,6,10,11]. By comparison, few researchers had studied the martensite at the FB region under the TEM. Up to now only Olden et al. [15] had observed a “martensite layer” that morphology was twin-like, in the DMW between AISI 8630 steel to Inconel 625/

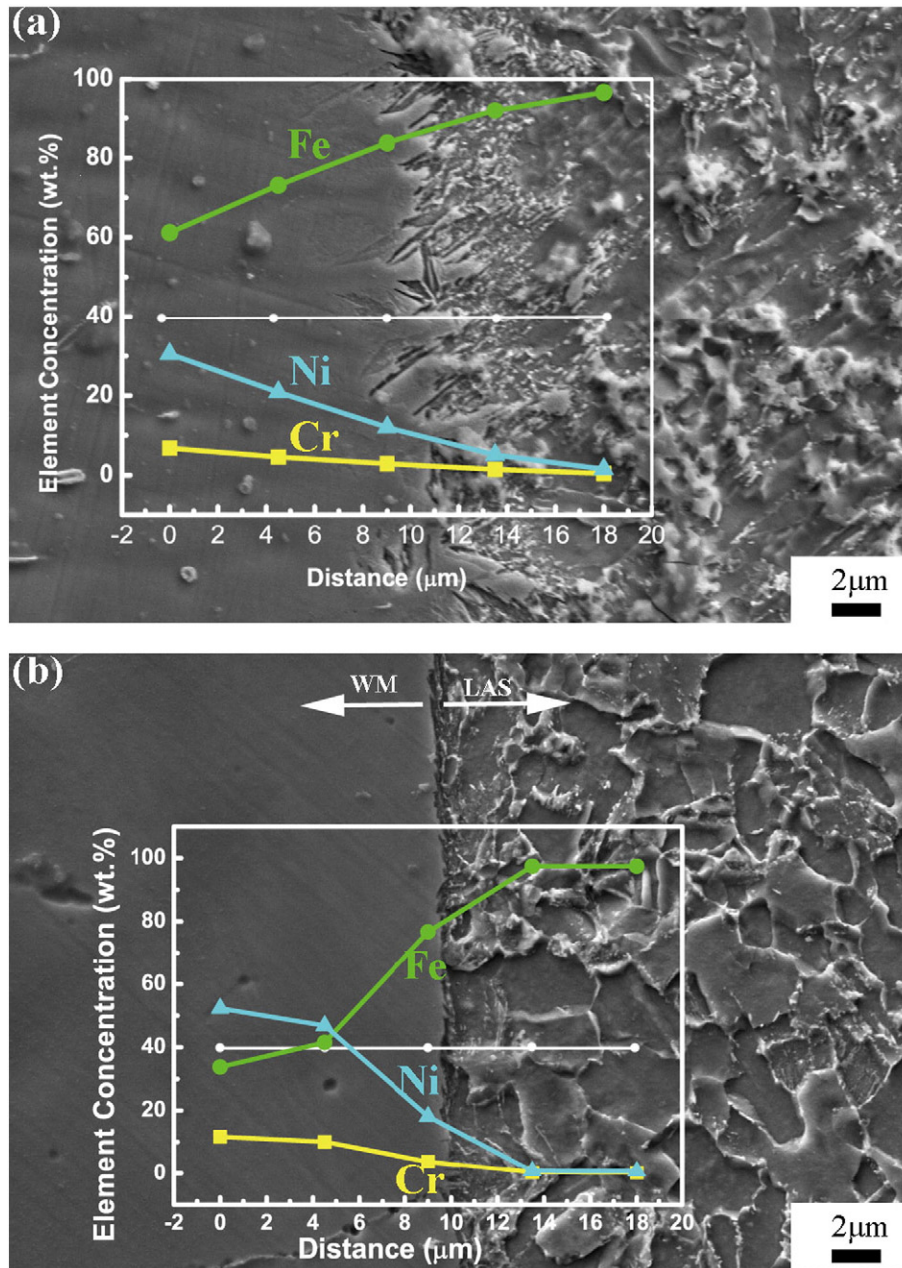


Fig. 4. SEM images of microstructure in (a) dark-etched [20] and (b) sharp FB zones, the elemental concentration across the FB are inserted.

725 weld metals using TEM. Several studies only found martensite-like structure under SEM at the FB region in DMWs between low alloy steel and the nickel-base WM [9,16,21], for example, Chung et al. [9] showed an SEM image of a hard “martensite-like” layer at the FB region in the DMW between LAS A508-III and Alloy 52. However, the crystal structure of the “martensite” in these studies was absent to be determined and the martensite layer was never found in the DMWs between A508-III and Alloy 82 in the TEM. In present study, a lenticular martensite layer, with a B.C.C structure (the same with α -ferrite), was found at the “martensite” FB, which was not determined in former studies on DMWs between LAS and nickel-based WM. It was found that the lenticular martensite in this study exhibited a K-S orientation relationship with the WM side, which indicated shear transformation of the F.C.C weld metal [22]. The formation mechanism of the martensite layer

had been discussed in our previous work, demonstrating that formation of the martensite was mainly determined by the distribution of alloy element [20]. Besides, the formation of the martensite can be proved by the Schaeffler diagram [23]. According to the EDS results shown in Fig. 4, the Ni_{eq} and the Cr_{eq} of the martensite FB and the sharp FB are plotted in the Schaeffler diagram in Fig. 9, respectively. It is clear that the martensite FB is in the martensite zone and the sharp FB is in the austenite zone of the Schaeffler diagram. It is consistent with the result of Wang et al. [21] who also predicted the formation of the martensite at the FB in a DMW between A508 and Alloy 52M. DuPont et al. [24] believed that the width of the martensite layer was controlled by the gradient of the element concentration across the FB. On the other hand, it was believed that for a filler metal containing enough nickel, the element concentration

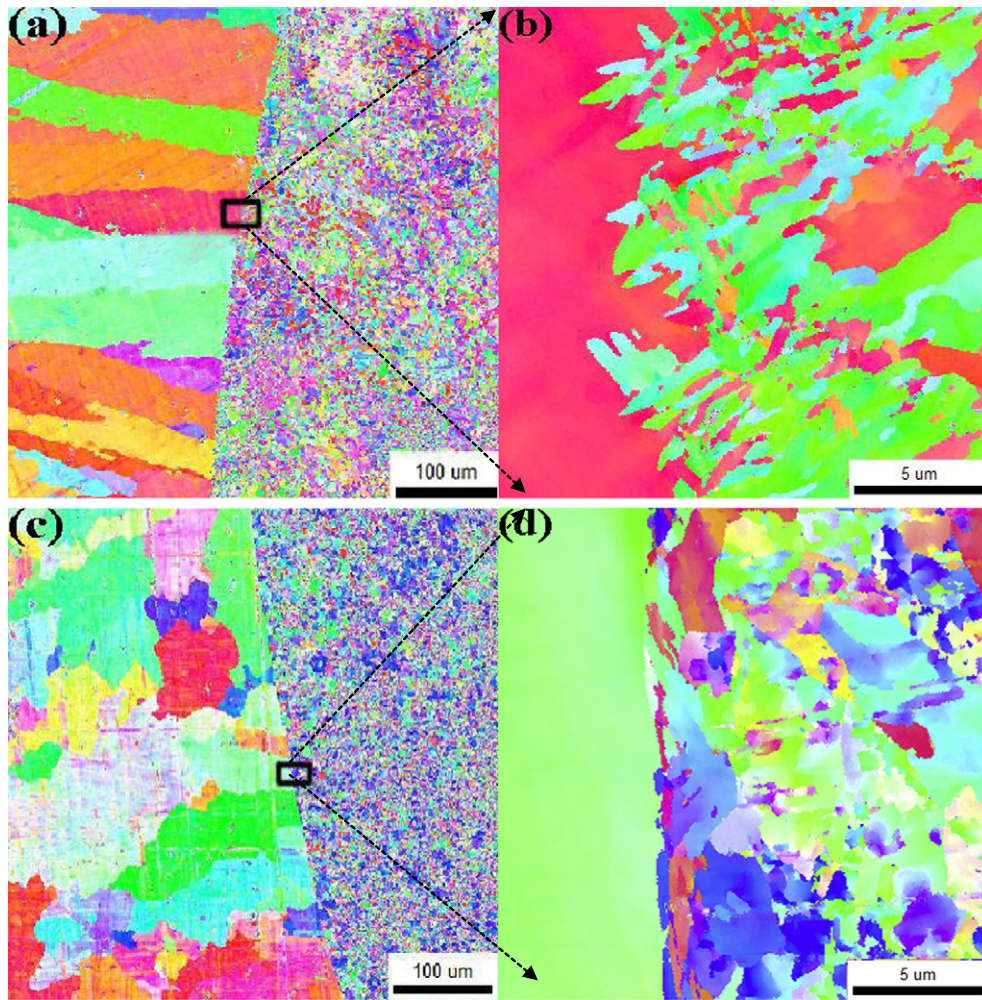


Fig. 5. IPF images of (a) martensite and (c) sharp FB regions, as well as the corresponding ((b) and (d)) high magnification images in the black rectangular boxes in (a) and (c), respectively.

gradient across the FB was large enough to produce a sharp FB without a martensite layer [25]. However, in this study, both the martensite FB and the sharp FB were found in the same FB, which was because the element concentration gradient across the martensite FB and the sharp FB were different, as shown in Fig. 4. Such a difference was believed to result from the different permanence time at high temperature during welding process. For example, a higher heating input, such as the heating input at the place between two welding beads (two times of heating), was expected to make more negative Gibbs free energy, which would lead to more activation of element's spontaneous diffusion and dilution between weld metal and base metal [26]. A longer permanence time would also lead to the growth of the LAS grains in the HAZ, which was consistent with the results.

It was apparent that in the LAS beside martensite FB, prior austenite grains were larger and HAZ was wider than those in the LAS beside sharp FB, as shown in Fig. 3, which indicated that the LAS beside the martensite FB experienced longer time at high temperature during welding. By comparison, for sharp FB, relatively shorter time at high temperature led to smaller austenitized LAS grains during welding. The close orientations of the HAZ grains beside martensite FB shown in Fig. 5 (b) indicated that they might belong to one prior austenite grain at high temperature during welding. While the HAZ grains beside the

sharp FB exhibited various orientations, indicating that they were in different prior austenite grains during welding. It also means that the prior austenite grains beside martensite FB were larger than those beside sharp FB, which was consistent with the OM observation shown in Fig. 3. These results indicated that the martensite FB experienced higher heat input than the sharp FB.

From Fig. 5 (a) and (c), it was found that the length/width ratio of the WM grains beside martensite FB was larger, indicating the cooling rate along the growing direction of the grains was high during the solidification of the WM. By comparison, the length/width ratio of the WM grains beside sharp FB was much lower, which indicated that the region beside the sharp FB did not have such a high cooling rate compared with the region beside the martensite FB. Since the martensite FB experienced higher heat input, the temperature at martensite FBs also might be higher than that at sharp FBs.

Former studies revealed that the maximum temperature, which could be attained at different points during different passes of the weld, could strongly affect the microstructure [27–29]. In this study, for martensite FB which might experience a higher heat input during welding, higher temperature and longer permanence time at high temperature would result in formation of a wide HAZ. At high temperature, austenitized LAS grains

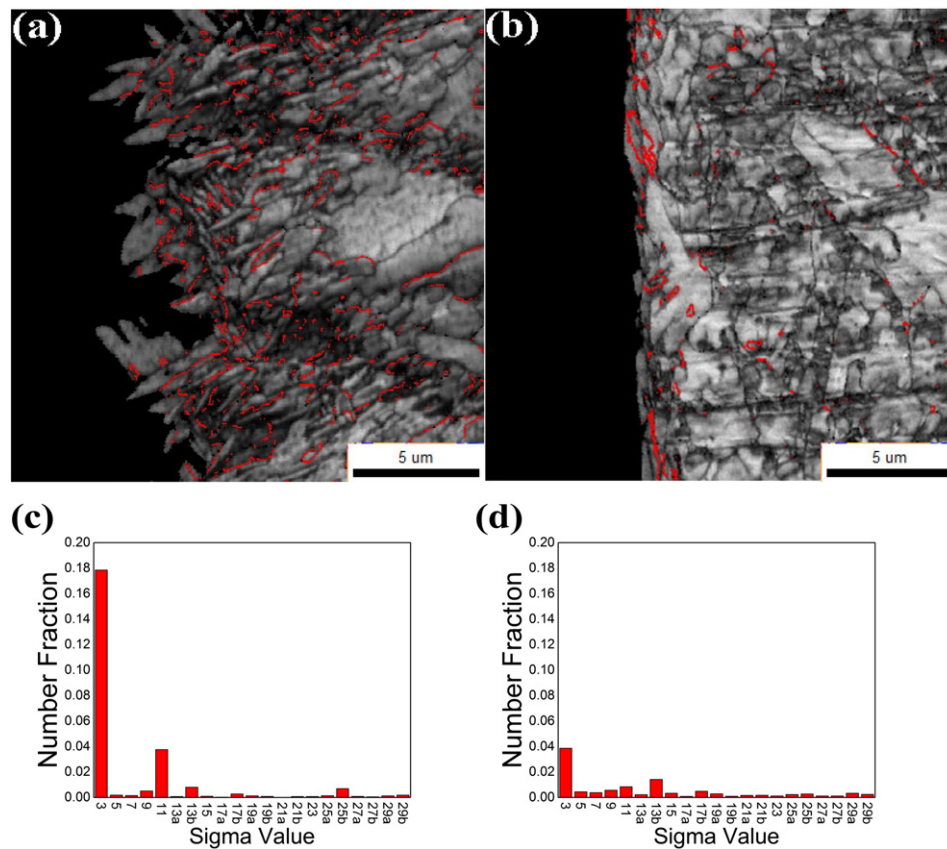


Fig. 6. EBSD results showing CSL $\Sigma 3$ boundaries of HAZ, and the corresponding statistic results for (a), (c) martensite and (b), (d) sharp FBs.

kept growing bigger and large prior austenite grains could be observed after cooling down. As the temperature was higher at the FB region, a larger temperature gradient would appear at the WM side, which was easy for the WM grains to grow along the temperature gradient direction and resulted in the solidification of columnar grains with high length/width ratios. The weld metal and the base metal would mix well at the FB region because of the higher temperature during welding and a gradual element concentration gradient would be found, as shown in Fig. 4 (a). Such a gradual element concentration gradient and the high cooling rate led to the formation of the martensite layer [20]. By comparison, the sharp FB experienced a lower heat input and the peak temperature at FB region during welding was relatively lower than the martensite FB. Therefore, the austenitized LAS grains would not grow as big as the grains beside the martensite FB and a narrower HAZ formed.

From Fig. 6, it was interesting to find that the amount of $\Sigma 3$ boundaries in HAZ beside martensite FB was much larger than that in HAZ beside the sharp FB. The $\Sigma 3$ boundaries usually are twin boundaries which may result from deformation. It indicated that the martensite FB region may have experienced more deformation process than the sharp FB region during welding. In welding process, plastic deformation usually occurs at the place where thermal stress reaches the yield strength of the metal [23, 27, 28]. As deformation would raise the M_s temperature, it is possible that the formation of martensite at the martensite FB was partially attributed to the deformation generated during welding [20].

The hardness profiles in Fig. 8 showed that the hardness peaks were located at both two FBs. However, the hardness of the martensite FB was much higher than that of the sharp FB, indicating martensite mainly contributed to the increase of the hardness at the FB. The relatively higher average hardness around the sharp FB in this study was attributed to the residual strain due to sharp change of the microstructure and chemical composition. The hardness profile across sharp FB was similar with former studies [5, 6, 10, 11], which believed that there were three reasons for the high hardness peak at the FB: (1) the residual strain generated during welding [5, 6]; (2) the carbides precipitated at the FB during welding [5, 6, 10, 11] and (3) some lath structures in the HAZ probably formed during cooling [5, 6]. In our study, there was no carbide and lath structure beside sharp FB, as shown in Figs. 4 (b), 5 (d), and 7 (c). Therefore, it was believed that the hardness peak at the sharp FB resulted from the residual strain generated during welding.

5. Conclusions

In this study, a DMW FB between low alloy steel A508-III and Alloy 82 weld metal was studied using OM, SEM with an EBSD system, TEM and a nano indentation. The main conclusions could be obtained as follows:

1. There were two types of FBs, martensite FB and sharp FB, along the same FB in the DMW between LAS A508-III and Alloy 82. The martensite FB had a martensite layer with a

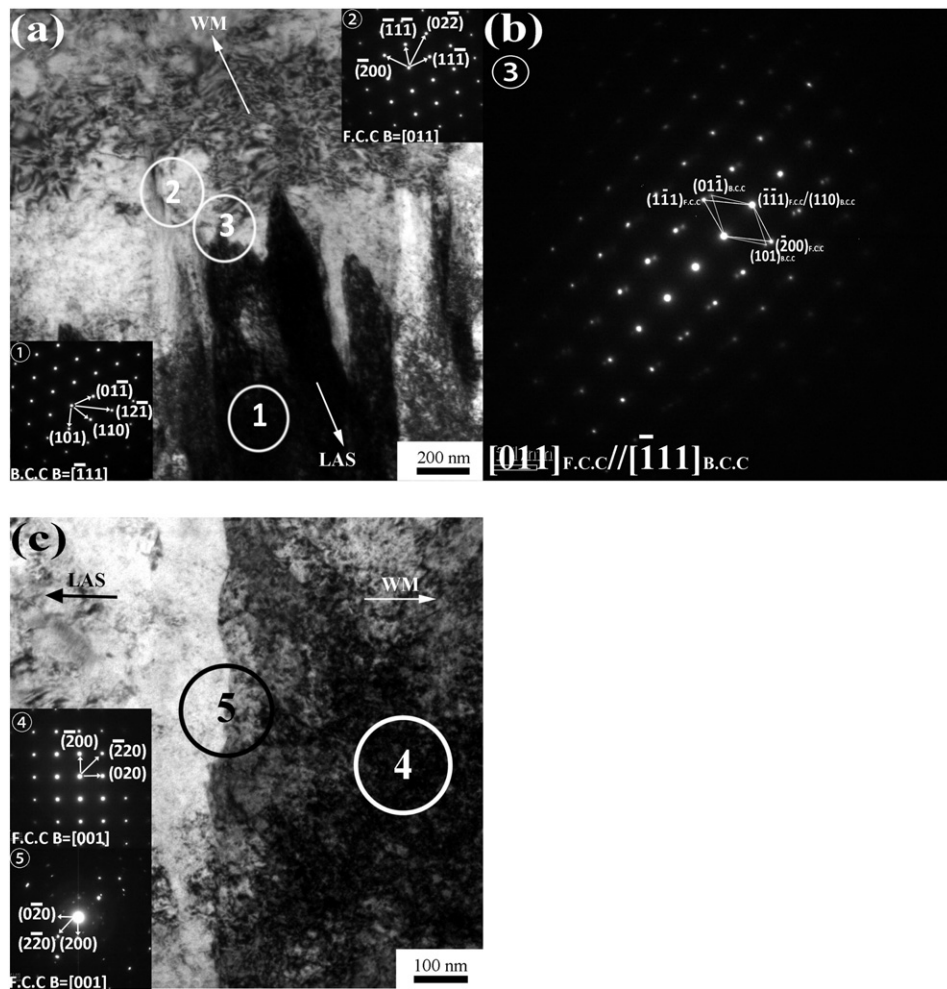


Fig. 7. TEM images of (a) the martensite and (c) the sharp FBs. The SAED patterns of locations 1, 2, 4, 5 are inserted, respectively, and the SAED pattern of location 3 is shown in panel (b).

width of dozens of microns and the sharp one exhibited a smooth boundary.

2. The type of the FB affected the chemical composition across the FB, and the morphology of the WM grains and the HAZ beside it. The

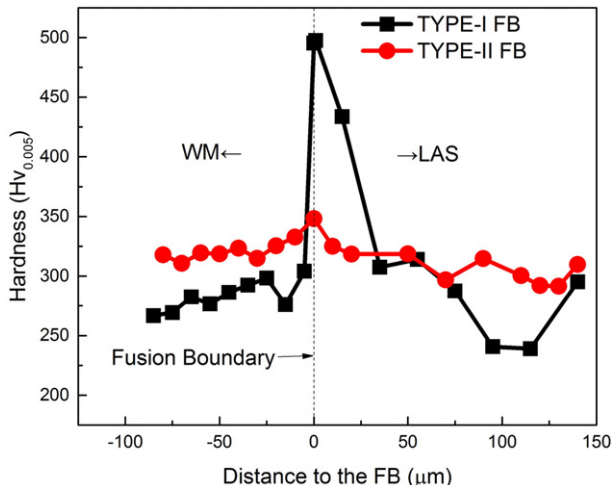


Fig. 8. Hardness profiles across the martensite and sharp FBs.

martensite FB region had a gradual change of elemental concentration across FB, columnar WM grains with high length/width ratios, a thick martensite layer and a wide HAZ with large prior austenite grains. By comparison, the sharp FB region had a relatively sharp change of elemental concentration across the FB, WM grains with low length/width ratios and a narrow HAZ with smaller prior austenite grains.

3. The type of the FB affected the distribution of CSL boundaries in the HAZ beside FB. Compared with the HAZ beside sharp FB, there were much more $\Sigma 3$ boundaries in the HAZ beside martensite FB.
4. The martensite at the martensite FB exhibited a K-S orientation relationship with the WM grains, while no orientation relationship was found between HAZ grains and WM grains at sharp FB.
5. The hardness maximum of the martensite FB was much higher than that of the sharp one, which resulted from the martensite layer at the martensite FB.

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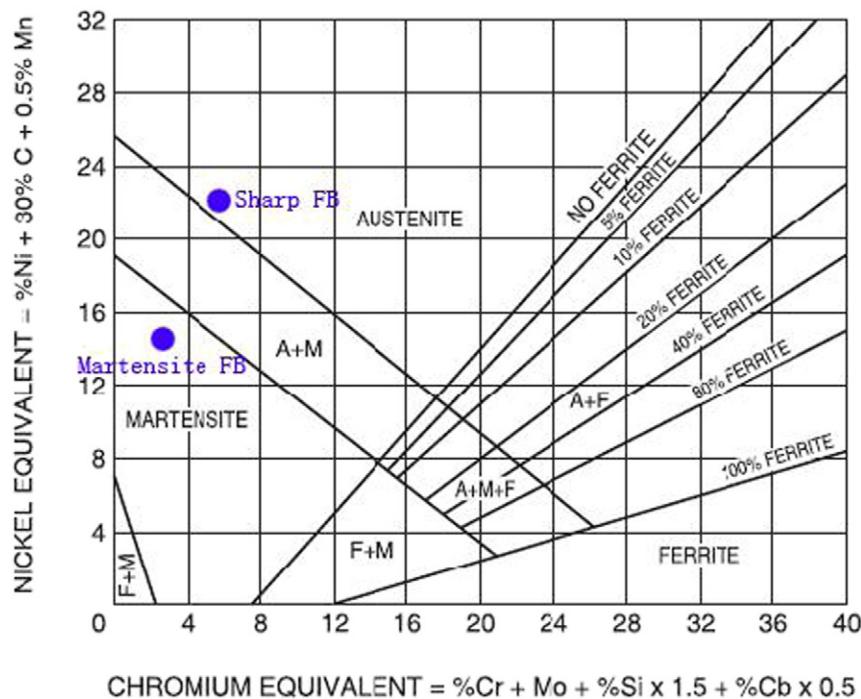


Fig. 9. Schaeffler diagram used for prediction of microstructure in ferritic, austenitic, and dissimilar welds. [23].

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